

Is Your Brain a Database?

When we think about how the human brain works, we often compare it to computers. It is an obvious analogy: the brain stores information, and a computer also stores information. However, some scientists argue that the similarity is deeper than just an analogy. Let's focus on how memory works in brains and databases. Does the human brain hold and recall memories like a structured database system, or is this just an analogy?

The answer to this question is important because it changes both our ideas about technology and our understanding of ourselves. The answer also changes how we do research in information technologies. If the brain truly functions like a database, then artificial intelligence systems could be designed to operate like human thinking. That means that large language models (LLMs) could become truly 'intelligent'. Artificial Intelligence could eventually think like a human by improving data storage, indexing, and query processing. The answer could also lead to developments in cybernetics and brain-computer interfaces. If computers and brains function the same way, then we could 'read' and 'write' information in the brain, like a database. If you want to know how to do karate, you could just download the knowledge into your brain, like in the movie, *The Matrix*. On the other hand, if memory functions differently in the brain and in a computer system, then new technologies like LLM models, brain-computer interfaces, and cybernetic implants are dead-ends. We will need completely new models to build intelligent systems.

How Does a Database Work?

A database is a system used to store and organize data, which at its most basic level is organized as bits (1s and 0s). Data in a database is arranged in a clear and logical structure. Unless the data is corrupted, it can be retrieved exactly as it was stored. For example, in a relational database, information is stored in tables, which look like spreadsheets. Each table has rows and columns. A row represents one item or record, and a column represents a type of information, such as a name, date, or number. This structure helps users understand and manage large amounts of data.

Databases also use rules, called schemas, to define how data should be organized. When users want to find information, they send queries, which are requests to search for specific data. The database system then looks through the stored data and returns the correct results. To make this process faster, databases often organize data in ways that make searching easier, so they do not need to check every record one-by-one. Large search engines, LLMs, and other advanced systems can handle large amounts of data at lightening fast speed, but at a basic level, they all follow this same structure. Overall, databases are designed to easily and quickly manage data, without changing or losing the data.

How does the human brain work?

The human brain works in a very different way. It is a biological system made of neurons connected by synapses. Instead of tables and schemas, the brain uses neural networks where information is stored in patterns of connections. Learning happens through changes in the strength of these connections. This process of making strong or weak connections is called 'plasticity'. Memory is not stored in a single location. Instead, memory is distributed across many neurons. The brain processes information in parallel, not in a logical and organized sequence like many database queries.

Human memory is based on experiences that we have in our lives. When you remember something, you are not ‘retrieving’ the information that is stored. Rather, you are re-experiencing the information. Because of plasticity, human memory is always changing. When you remember something, you reconstruct it based on experience, context, and associations, which are always changing, growing weaker or stronger. Emotions, the senses (taste, touch, smell, sight, and hearing), and knowledge all affect how your mind remembers information. This makes the brain more flexible but less accurate than a database system.

Brains and computers are essentially the same

When comparing the brain and databases, we can see both similarities and differences. On one side, cognitive scientists like Allen Newell and Herbert Simon suggest that human thinking can be understood as information processing, similar to how a computer executes algorithms. In this view, the brain stores data, processes it, and produces outputs, much like a Database Management System (DBMS) handling queries and transactions. Concepts like memory storage, retrieval, and indexing are basically the same as neural processes. Newell and Simon argue that, although the process is different in brains and computers, the functionality is essentially the same.

Brains and computers are essentially different

On the other side of the argument, many scientists and philosophers argue that the brain is fundamentally different from a computer. Researchers like Antonio Damasio emphasize that memory in the brain is based on lived experience and is deeply connected to the body and emotions. Unlike a database, which retrieves exact records, the brain reconstructs memories, and this process changes over time. Hubert Dreyfus argues that human intelligence cannot be reduced to symbolic data processing, as it depends on context, intuition, and the human body. Dreyfus says the brain is not just a storage system, but a dynamic, adaptive network that cannot be fully described using database concepts like schemas, queries, or transactions. Thinking, especially critical thinking, is not just managing data.

What do you think?

No one is arguing that the brain “is” a computer. Rather, the question is, if they are similar, *how* similar are they. The comparison could just be a useful analogy, or something more insightful about how computers and brains really work. Whatever conclusion you come to, trying to answer this question helps us better understand both human memory and computer information processing. So, what do you think? Does your brain basically function like a database when you remember something? If so, how similar are they? Or, do you think brains and computers function in completely different ways?